

Content

The NOKOV SDK user manual is used to help developers understand the usage rules of the SDK. This manual includes all interfaces and detailed instructions, providing users with the ultimate goal of receiving motion capture data from the NOKOV program.

Applicable NOKOV programs and versions:

XINGYING 1.0.0.4321 and above;
XING 1.4.0.4537 and above;

一、type definition

1、sRigidBodyData

Rigid Info

```
public struct sRigidBodyData {  
    /// <summary> ///  
    The identifier ///  
    </summary>  
    public Int32 Id; ///  
    <summary> ///  
    The x///  
    </summary>  
    public float X; ///  
    <summary> ///  
    The y  
    /// </summary>  
    public float Y; ///  
    <summary> ///  
    The z  
    /// </summary>  
    public float Z; ///  
    <summary> ///  
    The qx  
    /// </summary>  
    public float QX;  
    /// <summary> ///  
    The qy  
    /// </summary>  
    public float QY;
```

```
/// <summary> /// The qz
/// </summary> public
float QZ; /// <summary> ///
The qw /// </summary>
public float QW; ///
<summary> /// The n
markers /// </summary>
public int NMarkers; ///
<summary> /// The
markers /// </summary>
public IntPtr Markers; ///
<summary> /// The marker
ids /// </summary> public
IntPtr MarkerIDs; ///
<summary> /// The marker
sizes /// </summary>
public IntPtr MarkerSizes;
/// <summary> /// The
mean error /// </summary>
public float MeanError; ///
<summary> /// The
parameters /// </summary>
public Int16 Params;
```

```
}
```

Members

ID //rigid ID

x, y, z rigid //position

qx, qy, qz, qw //rigid rotation

NMarkers

The Marker count

Markers

Marker data

MarkerIDs

Marker ID

MarkerSizes

Marker size

MeanError

deviation

params

reserved

2、 sSkeletonData

Skeleton Info.

```
public struct sSkeletonData {  
    /// <summary>  
    /// The identifier  
    /// </summary>  
    public Int32 Id;  
    /// <summary>  
    /// The rigid body count  
    /// </summary>  
    public Int32 RigidBodyCount;  
    /// <summary>  
    /// The rigid bodies  
    /// </summary>  
    public IntPtr RigidBodies; // Pointer to sRigidBodyData[RigidBodyCount]  
}
```

Members

ID
Body ID
RigidBodyes
Skeleton count
sRigidBodyData
Skeleton Data Info

3、sMarker

Marker info。

```
public struct sMarker {  
    /// <summary> ///  
    The identifier ///  
    </summary>  
    public Int32 Id; ///  
    <summary> ///  
    The position x ///  
    </summary>  
    public float X; ///  
    <summary> ///  
    The position y ///  
    </summary>  
    public float Y; ///  
    <summary> ///  
    The position z ///  
    </summary>  
    public float Z; ///  
    <summary> ///  
    The size  
    /// </summary> public float  
    Size; /// <summary> /// The  
    parameters(reserved) ///  
    </summary> public Int16  
    Params;  
}
```

Members

ID Marker ID

x, y, z Marker position

size Marker size

params
reserved

4、sFrameOfMocapData

One Frame data of mocap。

```
public struct sFrameOfMocapData {  
    /// <summary> /// The frame  
    number /// </summary>  
    public Int32 FrameNumber;  
  
    /// <summary>  
    /// The marker set count  
    /// </summary>  
    public Int32 MarkerSetCount;  
    /// <summary>  
    /// The marker sets  
    /// </summary>  
    [MarshalAs(UnmanagedType.ByValArray, SizeConst =  
NokovSDKConstants.MaxModels)] public  
    sMarkerSetData[] MarkerSets;  
  
    /// <summary>  
    /// The undefined marker count,  
    /// </summary>  
    public Int32 OtherMarkerCount;  
    /// <summary>  
    /// The undefined markers, Pointer to float[OtherMarkerCount][3] ///  
    </summary>  
    public IntPtr OtherMarkers;
```

```
/// <summary>
/// The rigid body count
/// </summary>
public Int32 RigidBodyCount;
/// <summary>
/// The rigid bodies
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxRigidBodies)] public
sRigidBodyData[] RigidBodies;

/// <summary>
/// The skeleton count
/// </summary>
public Int32 SkeletonCount;
/// <summary>
/// The skeletons
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxSkeletons)]
public sSkeletonData[] Skeletons;

/// <summary>
/// The labeled marker count
/// </summary>
public Int32 LabeledMarkerCount;
/// <summary>
/// The labeled markers
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxLabeledMarkers)]
public sMarker[] LabeledMarkers;

/// <summary>
/// The analogdata count
/// </summary>
public Int32 AnalogdataCount;
/// <summary>
/// The analogdata
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxAnalogChannels)]
public float[] Analogdata;
```

```
/// <summary>
/// The host defined time delta between capture and send
/// </summary>
public float FLatency;
/// <summary>
/// The SMPTE timecode (if available)
/// </summary>
public UInt32 Timecode;
/// <summary>
/// The timecode subframe
/// </summary>
public UInt32 TimecodeSubframe;
/// <summary>
/// The FrameGroup timestamp, the number of milliseconds since the Epoch
1970-01-01 00:00:00 +0000 (UTC).
/// </summary> public
UInt64 Timestamp; ///
<summary>
/// The host defined parameters
/// </summary>
public Int16 Params; }
```

Members

FrameNumber
frame index

MarkerSetsCount
MarkerSet count

MocapData MarkerSet Info

OtherMarkersCount
Unnamed Marker count

OtherMarkers
Unnamed Marker data

RigidBodiesCount
rigid count

RigidBodies

Rigid info

SkeletonsCount
body/Skeletons
count

Skeletons body/rigid
info

LabeledMarkersCount
named Marker count

LabeledMarkers
named Marker data

Analogdatas
Analog Channel count

Analogdata
Analog data

FLatency
Latency

Timecode SMPTE
Timecode

TimecodeSubframe
sub frame of the frame in timecode

TimeStamp
(ms)

params
reserved

6、 Verbosity

Log level


```
public enum NokovVerbosity
{
    /// <summary>

    /// The verbosity none

    /// </summary>

    Verbosity_None = 0,

    /// <summary>

    /// The verbosity information

    /// </summary>

    Verbosity_Info, ///

    <summary> /// The verbosity

    warning /// </summary>

    Verbosity_Warning, ///

    <summary> /// The verbosity

    error /// </summary>

    Verbosity_Error,
```

```
/// <summary>

/// The verbosity debug

/// </summary>

Verbosity_Debug,

}
```

7、sMarkerSetDescription

MarkerSet Description。

```
public struct sMarkerSetDescription {
    /// <summary> /// The
    MarkerSet name ///
    </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
    public string Name;
    /// <summary>
    /// The number of markers in MarkerSet
    /// </summary>
    public Int32 MarkerCount;
    /// <summary>
    /// The marker names, char**, "array of marker names" ///
    </summary>
    public IntPtr MarkerNames;
}
```

Members

Name
MarkerSet name

MarkersCount
Marker count

MarkerNames
Marker name array

8、sRigidBodyDescription

Rigid Description。

```
public struct sRigidBodyDescription
{
    /// <summary>
    /// The RigidBody name
    /// </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
    public string Name;

    /// <summary>
    /// The RigidBody identifier
    /// </summary>
    public Int32 Id;
    /// <summary>
    /// ID of parent Rigid Body (in case hierarchy exists) ///
    </summary>
    public Int32 ParentId;
    /// <summary>
    /// The offset position x relative to parent ///
    </summary>
    public float OffsetX;
    /// <summary>
    /// The offset position y relative to parent ///
    </summary>
    public float OffsetY;
    /// <summary>
    /// The offset position z relative to parent ///
    </summary>
    public float OffsetZ;
    /// <summary>
    /// The Orientation qx relative to parent ///
    </summary>
    public float Qx;//
    /// <summary>
    /// The Orientation qy relative to parent
```

```
    /// </summary>
    public float Qy;
    /// <summary>
    /// The Orientation qz relative to parent ///
    </summary>
    public float Qz;
    /// <summary>
    /// The Orientation qz relative to parent ///
    </summary>
    public float Qw;
}
```

Members

- Name
Rigid name
- ID
Rigid ID
- parentID
parent rigid ID
- offsetx, offsety, offsetz
xyz offset to parent
- qx, qy, qz, qw
rotation to parent

9、sSkeletonDescription
Skeleton description。

```
public struct sSkeletonDescription
{
    /// <summary>
    /// The name
    /// </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
```

```
public string Name;

/// <summary> /// The
identifier /// </summary>
public Int32 Id; /// <summary>
/// The rigid body count ///
</summary> public Int32
RigidBodyCount;

/// <summary>
/// The rigid bodies
/// </summary>
[MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxSkeletonRigidBodies)]
public sRigidBodyDescription[] RigidBodies;
}
```

Members

Name
Body name
ID
Body ID
RigidBodiesCount
The rigid count of the body
RigidBodies
rigids array

10、 sDataDescription

```
data description。

public struct sDataDescription {
    /// <summary> /// The
description type
```

```
    /// </summary> public Int32
    DescriptionType; ///
    <summary> /// The description
    pointer /// </summary> public
    IntPtr Description;
}
```

Members

type
DescriptionType

11、sDataDescriptions

```
public struct sDataDescriptions {
    /// <summary>
    /// The data description count
    /// </summary>
    public Int32 DataDescriptionCount;

    /// <summary>
    /// The data descriptions
    /// </summary>
    [MarshalAs(UnmanagedType.ByValArray, SizeConst =
NokovSDKConstants.MaxModels)]
    public sDataDescription[] DataDescriptions;
}
```

Members

sDataDescriptions
description count

DataDescriptions
description array

12、sMarkerSetData

MarkerSet info。

```
public struct sMarkerSetData
{
    /// <summary>
    /// The name
    /// </summary>
    [MarshalAs(UnmanagedType.ByValTStr, SizeConst =
NokovSDKConstants.MaxNameLength)]
    public string Name;

    /// <summary>
    /// The marker count
    /// </summary>
    public Int32 MarkerCount;
    /// <summary>
    /// The markers, Pointer to float[MarkerCount][3]
    /// </summary>
    public IntPtr Markers;
}
```

Members

- Name
MarkerSet name
- MarkersCount
Marker count
- Markers
Marker data

二、API

1、Initialize

Prerequisite for initializing SDK and calling other SDK functions。
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]

```
public static extern NokovSDKError Initialize(IntPtr clientHandle, string
szServerAddress);
```

Parameters

szServerAddress [in]

the ip of server(mocap software)

2、 Uninitialize

Release SDK resources and call before the program ends.

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError Uninitialize(IntPtr clientHandle);
```

3、 SetDataCallback

Set capture data callback。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError SetDataCallback(IntPtr clientHandle,
NokovSDKFrameReceivedCallback dataCallback, IntPtr pUserData);
```

Parameters dataCallback [in]

callback func.

pUserData [in]

user data

Callback Function Parameters

pUserData [out]

user data

4、 SetMessageCallback

set log callback。

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError SetMessageCallback(IntPtr
clientHandle, NokovSDKMessageCallback messageCallBack);
```


);

Parameters messageCallBack [in]

mocap message callback

5、GetServerDescription

Get mocap system description info.

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =  
NokovSDKConstants.NokovSDKLibCallingConvention)]
```

```
    public static extern NokovSDKError GetServerDescription(IntPtr  
pServerDescription);
```

Parameters

pServerDescription [out]

mocap system application description.

6、GetDataDescriptions

Get mocap data description.

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =  
NokovSDKConstants.NokovSDKLibCallingConvention)]
```

```
    public static extern NokovSDKError GetDataDescriptions(IntPtr  
clientHandle, out IntPtr pDataDescriptions);
```

Parameters

pDataDescriptions [out]

mocap data description.

7、DecodeTimecode

Decode the timecode.

```
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =  
NokovSDKConstants.NokovSDKLibCallingConvention)]
```

```
    public static extern NokovSDKError DecodeTimecode(IntPtr clientHandle, uint  
inTimecode, uint inTimecodeSubframe, ref int hour, ref int minute, ref int second, ref int  
frame, ref int subframe);
```

Parameters

inTimecode [in]
timecode

inTimecodeSubframe [in]

timecode

hour

[out]

minute

[out]

second

[out]

frame

[out] [1-30]

subframe

[out] sub frame

Return Values

TRUE success, FALSE fail.

8、TimecodeStringify

Format the string after decoding the time code.
[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention =
NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern NokovSDKError TimecodeStringify(IntPtr clientHandle,
uint inTimecode, uint inTimecodeSubframe, string Buffer, int BufferSize);

Parameters

inTimecode
[in]

inTimecodeSubframe
[in]

Buffer [out]

BufferSize [in]

Return Values

TRUE success, FALSE fail.

9、NokovFreeFrame

Free the frame data buffer。

[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention = NokovSDKConstants.NokovSDKLibCallingConvention)]

public static extern NokovSDKError NokovFreeFrame(IntPtr clientHandle, IntPtr pFrameOfMocapData);

Parameters

pFrameOfMocapData
[in] frame data

10、GetLastFrameOfMocapData

Get one frame of mocap data。

[DllImport(NokovSDKConstants.NokovSDKLibDllBaseName, CallingConvention = NokovSDKConstants.NokovSDKLibCallingConvention)]
public static extern IntPtr GetLastFrameOfMocapData(IntPtr clientHandle);

Return Values

One frame, NULL fail.

Remarks

The frame should free use [NokovFreeFrame](#) after used. it non thread safe.